

Mega-Cap Technology Drawdown and Accumulation Framework

A Historical Study of Long-Term Returns, Drawdown Frequency, Recovery Characteristics, and Staged Buying Strategies

Executive Summary

Long-term investing in mega-cap technology companies is often discussed primarily in terms of return. Investors compare compound annual growth rates, identify the strongest historical winners, and then assume that the companies with the highest long-term returns should also receive the largest allocations.

That approach is incomplete.

The path through which returns are generated matters almost as much as the final return itself. A stock that compounds at more than 40% annually while repeatedly experiencing 30%–50% drawdowns requires a fundamentally different investment process from a stock that compounds at approximately 20% while typically experiencing 15%–20% annual drawdowns.

More importantly, the same percentage decline does not carry the same information across different stocks.

A 30% drawdown in Microsoft has historically been extremely unusual. A 30% drawdown in NVIDIA has occurred much more frequently. For Tesla, a 30% drawdown has historically been close to a routine feature of the stock rather than an exceptional event.

This distinction has direct implications for portfolio construction and staged buying.

This report analyzes nine large technology and technology-adjacent companies:

- NVIDIA
- Tesla
- Broadcom
- Meta Platforms
- Apple
- Taiwan Semiconductor Manufacturing Company
- Amazon
- Microsoft
- Alphabet

The analysis focuses primarily on complete calendar years from 2010 through 2025, using adjusted daily prices that account for stock splits and cash distributions. Meta is evaluated from its post-IPO history, while

Tesla's incomplete 2010 IPO year is excluded from the core annual comparisons. 2026 year-to-date observations are also excluded from the long-term annual statistics.

The framework evaluates each company across several dimensions:

- Compound annual growth rate
- Percentage of positive calendar years
- Median annual maximum drawdown
- Annualized volatility
- Downside volatility
- Typical worst one-month return
- Typical worst three-month return
- Median recovery time
- Frequency of 20%, 30%, 40%, and 50% annual drawdowns

Several conclusions stand out.

First, **Microsoft, Alphabet, and Apple are the strongest candidates for continuous long-term dollar-cost averaging.** Their typical drawdowns are relatively moderate, and deep drawdowns occur infrequently enough that waiting for a major crash can lead to prolonged underinvestment.

Second, **Broadcom, Meta, and Amazon are better suited to a hybrid strategy combining ongoing investment with dry powder reserved for larger corrections.** These companies have strong long-term return profiles but offer more frequent opportunities to increase exposure during deeper drawdowns.

Third, **TSMC should be treated as a cyclical compounder rather than a purely mechanical dollar-cost-averaging asset.** Its historical price behavior is more stable than many semiconductor stocks, but investment decisions during major drawdowns should still incorporate semiconductor demand, capacity utilization, capital expenditure cycles, and geopolitical risk.

Fourth, **NVIDIA and Tesla require a different framework entirely.** A 20% decline is not sufficiently unusual to justify aggressive buying based on price alone. For NVIDIA, the 30%–40% range is more meaningful. For Tesla, even a 30% decline has historically been common enough that major incremental capital should generally be reserved for deeper corrections.

The central principle of the report is therefore simple:

A drawdown should be judged relative to the historical behavior of the specific stock, not against a universal percentage threshold.

The objective is not to predict exact bottoms. It is to systematically increase investment intensity when a high-quality company experiences a drawdown that is unusually deep relative to its own historical distribution while its long-term investment thesis remains intact.

1. Methodology

1.1 Sample and Data

The study covers the following companies:

Company	Ticker
NVIDIA	NVDA
Tesla	TSLA
Broadcom	AVGO
Meta Platforms	META
Apple	AAPL
Taiwan Semiconductor Manufacturing	TSM
Amazon	AMZN
Microsoft	MSFT
Alphabet	GOOGL

The primary comparison period is 2010–2025.

The following adjustments are made for comparability:

- Meta is evaluated from its public-market history beginning in 2012.
- Tesla's partial 2010 IPO year is excluded from the core full-year comparison.
- 2026 is excluded because the year is incomplete.
- Adjusted Close is used so that stock splits and cash distributions are reflected in the return series.

The purpose of the dataset is not to forecast future returns directly. Instead, it is used to estimate the historical behavior of each stock so that an investor can distinguish between ordinary volatility and genuinely unusual drawdowns.

1.2 Compound Annual Growth Rate

Compound annual growth rate measures the annualized rate at which an investment would have grown between the beginning and end of the observation period.

The calculation is:

$$\text{CAGR} = (\text{Ending Price} / \text{Starting Price})^{(1 / \text{Number of Years})} - 1$$

CAGR is useful for measuring long-term wealth creation.

Its main limitation is that it ignores the path of returns.

Two stocks can produce similar long-term CAGR figures while delivering completely different investment experiences. One may compound steadily, while another may alternate between extraordinary rallies and severe multi-year drawdowns.

For this reason, CAGR should always be evaluated alongside drawdown and recovery metrics.

1.3 Positive-Year Ratio

The positive-year ratio measures the percentage of complete calendar years in which a company's adjusted share price finished above its starting level.

The calculation is:

Positive-Year Ratio = Number of Positive Calendar Years / Total Complete Calendar Years

This metric provides a simple measure of annual return consistency.

It should not be interpreted as a risk measure by itself. A stock may finish a year in positive territory while still experiencing a severe drawdown during that year.

1.4 Annual Maximum Drawdown

Annual maximum drawdown measures the largest peak-to-trough decline experienced within a calendar year.

For each trading day, the stock's price is compared with the highest price previously reached during that same calendar year.

For example:

A stock begins the year at \$100, rises to \$150, falls to \$105, and finishes at \$140.

Its calendar-year return remains positive, but the investor experienced a 30% decline from the \$150 peak to the \$105 trough.

The annual return therefore does not fully describe the investment experience.

In this report, the **median annual maximum drawdown** is used as an estimate of a company's typical annual drawdown.

Median is preferred to average because a small number of extreme years can materially distort the mean.

1.5 Annualized Volatility

Annualized volatility measures the dispersion of daily returns.

It is calculated from the standard deviation of daily returns and annualized using approximately 252 trading days.

In simplified form:

Annualized Volatility = Daily Return Volatility × Square Root of 252

Higher volatility indicates a wider distribution of daily price movements.

However, conventional volatility treats upside and downside movements symmetrically.

A 5% rally increases measured volatility just as a 5% decline does.

For a long-term investor, downside volatility is therefore also useful.

1.6 Downside Volatility

Downside volatility focuses specifically on negative-return observations.

It provides a more intuitive estimate of how violently a stock tends to move during declining periods.

A company can have high total volatility partly because of strong upside moves. Downside volatility isolates the portion of volatility that is more directly associated with investor discomfort and capital loss.

1.7 Worst One-Month and Three-Month Returns

The study also measures a typical year's worst rolling one-month and three-month return.

These statistics answer a practical question:

If an investor happened to buy near a short-term high, how much could the position typically lose over the following one to three months?

This provides a more realistic view of short-term holding difficulty than annual volatility alone.

For example, Microsoft's typical worst one-month return has historically been approximately -10.6%, while Tesla's has been approximately -27.0%.

That difference has significant implications for position sizing and staged buying.

1.8 Recovery Time

Recovery time measures the number of trading days from a previous price peak until the stock eventually returns to that peak.

It captures the entire underwater period:

Peak → Decline → Trough → Recovery

This metric is particularly useful because drawdown depth alone does not fully describe investor experience.

Two stocks may both decline 40%, but one may recover within several months while the other may take several years.

Long recovery periods increase both the psychological and opportunity cost of holding the position.

1.9 Drawdown Recurrence

The final major risk metric is drawdown recurrence.

For each company, the study measures how frequently a calendar year contains a maximum drawdown of at least:

- 10%
- 20%
- 30%
- 40%
- 50%

The calculation is:

Drawdown Recurrence Interval = Observation Years / Number of Years Meeting the Threshold

For example, if a company has 16 complete years of data and eight of those years experienced at least a 30% maximum drawdown:

16 divided by 8 equals 2 years.

This means that approximately one out of every two historical calendar years contained a drawdown of at least 30%.

It does not mean that the stock mechanically falls 30% exactly once every two years.

The recurrence interval should instead be interpreted as a measure of historical rarity.

2. Return vs Drawdown Analysis

2.1 Long-Term Return and Risk Summary

Company	CAGR	Positive Years	Median Max Drawdown	Annualized Volatility	Downside Volatility	Typical Worst 1M	Typical Worst 3M	Median Recovery
NVDA	46.2%	68.8%	-28.7%	46.0%	27.9%	-17.7%	-15.1%	89.5 days
TSLA	44.6%	86.7%	-35.7%	54.1%	36.5%	-27.0%	-24.4%	204 days
AVGO	41.9%	93.8%	-22.9%	34.7%	24.6%	-19.0%	-14.6%	89.5 days
META	28.1%	84.6%	-19.7%	36.5%	24.3%	-15.3%	-14.0%	78 days
AAPL	26.5%	81.3%	-18.8%	26.5%	19.0%	-16.8%	-14.5%	100.5 days
TSM	26.4%	87.5%	-20.3%	26.8%	19.0%	-16.2%	-16.8%	130.5 days
AMZN	24.7%	81.3%	-21.2%	32.4%	21.2%	-17.6%	-13.7%	107 days
MSFT	21.0%	81.3%	-16.0%	23.1%	16.7%	-10.6%	-11.4%	80.5 days
GOOGL	20.7%	75.0%	-19.1%	28.0%	18.4%	-14.1%	-14.3%	121.5 days

2.2 Stable Compounders

Microsoft stands out as the lowest-risk historical compounder in the group.

Its 21.0% historical CAGR is lower than that of NVIDIA, Broadcom, or Tesla, but its median annual maximum drawdown of only 16.0% is the shallowest in the sample.

Microsoft also has:

- The lowest typical annualized volatility
- The lowest downside volatility
- The mildest typical worst one-month return
- A relatively short recovery period

Its return path has therefore historically required substantially less tolerance for volatility.

Apple also displays a highly attractive return-risk profile.

Its historical CAGR of 26.5% is materially above Microsoft's, while its typical annual drawdown remains below 20%.

Alphabet's historical CAGR is lower, but its drawdown characteristics remain broadly consistent with a high-quality platform compounder rather than a high-beta cyclical asset.

These companies have an important strategic implication:

The investor should generally prioritize staying invested over waiting for unusually deep corrections.

2.3 High-Return Compounders

Broadcom and NVIDIA produced the strongest historical long-term return profiles outside Tesla.

Broadcom is particularly notable.

Its historical CAGR is approximately 41.9%, while 93.8% of complete calendar years were positive.

Despite that return profile, its median annual maximum drawdown was approximately 22.9%, substantially below NVIDIA and Tesla.

Broadcom therefore delivered an unusually strong combination of:

- High CAGR
- High positive-year consistency
- Moderate typical drawdown relative to return
- Fast recovery

NVIDIA delivered an even higher CAGR of approximately 46.2%, but investors had to tolerate significantly more volatility.

Its median annual maximum drawdown approached 29%, and annualized volatility was approximately 46%.

NVIDIA therefore represents a strong historical example of a **high-beta compounder**: extraordinary long-term wealth creation combined with unusually deep and frequent intermediate drawdowns.

2.4 Tesla: High Return with Extreme Holding Difficulty

Tesla's historical CAGR is also extraordinary at approximately 44.6%.

However, virtually every major risk metric is materially worse than the rest of the group.

Its typical annual maximum drawdown was approximately 35.7%.

Its annualized volatility exceeded 54%.

Its downside volatility was approximately 36.5%.

Its typical worst one-month decline was approximately 27%.

Its median recovery period was more than 200 trading days.

This means that Tesla's strong long-term historical return should not be interpreted as evidence that it is an easy asset to hold.

The return came with unusually high path dependency.

Investors who allocate too aggressively after relatively small corrections risk exhausting capital long before a full drawdown develops.

2.5 Meta: Moderate Typical Risk, Material Tail Risk

Meta's median annual maximum drawdown of approximately 19.7% appears relatively benign.

It is broadly comparable with Alphabet and Apple.

However, Meta's distribution contains substantially more severe tail events.

Its risk profile can therefore be described as:

Moderate normal-year volatility with occasional severe valuation resets.

Meta does not behave like Tesla, where large drawdowns are routine.

Instead, Meta can trade relatively normally for extended periods and then experience a dramatic repricing when the market begins questioning:

- User engagement
- Platform relevance
- Advertising economics
- Capital allocation
- Competitive positioning
- Technology transitions

This makes deep drawdowns potentially attractive, but also increases the importance of fundamental thesis validation.

2.6 TSMC: A Cyclical Company with Surprisingly Stable Price Characteristics

TSMC's historical risk profile is more stable than might be expected from a semiconductor company.

Its historical CAGR of approximately 26.4% is almost identical to Apple's.

Its median annualized volatility of approximately 26.8% is also close to Apple.

Its median annual maximum drawdown was around 20.3%.

This suggests that TSMC should not be treated in the same way as a more commodity-like memory semiconductor company.

However, the tail risks are different.

TSMC remains exposed to:

- Semiconductor demand cycles
- Foundry utilization
- Customer capital expenditure
- Advanced-node demand
- Industry inventory cycles
- Geopolitical risk

Its investment framework should therefore combine historical price statistics with industry-cycle analysis.

2.7 Return Efficiency

A simple, non-standard way to compare historical return against typical drawdown is to divide CAGR by the absolute value of median annual maximum drawdown.

This is not a conventional portfolio-management ratio and should not replace Sharpe or Sortino analysis.

It is simply a useful heuristic for comparing how much long-term historical return was generated relative to typical annual drawdown.

Rank	Company	CAGR / Median Annual Max Drawdown
1	AVGO	1.83
2	NVDA	1.61
3	META	1.42
4	AAPL	1.41
5	MSFT	1.31
6	TSM	1.30
7	TSLA	1.25
8	AMZN	1.17
9	GOOGL	1.08

Broadcom stands out in this historical comparison.

NVIDIA also produced exceptional compensation for its typical drawdown risk.

Tesla's long-term return was extraordinary, but the magnitude of its typical drawdown reduced its relative efficiency.

The purpose of this comparison is not to forecast future rankings. It simply highlights the very different paths through which similar long-term wealth creation can occur.

3. Drawdown Frequency Charts

3.1 Historical Drawdown Recurrence

Company	≥10%	≥20%	≥30%	≥40%	≥50%
NVDA	1.07 years	1.45 years	2.0 years	4.0 years	4.0 years
AAPL	1.07 years	2.29 years	4.0 years	Not observed	Not observed
GOOGL	1.23 years	2.29 years	5.33 years	16 years	Not observed
MSFT	1.23 years	4.0 years	16 years	Not observed	Not observed

Company	≥10%	≥20%	≥30%	≥40%	≥50%
AMZN	1.0 year	2.0 years	5.33 years	16 years	16 years
TSM	1.0 year	2.0 years	8.0 years	16 years	16 years
AVGO	1.07 years	1.45 years	4.0 years	8.0 years	Not observed
META	1.08 years	2.17 years	3.25 years	6.5 years	13 years
TSLA	1.0 year	1.0 year	1.25 years	3.0 years	7.5 years

3.2 Annual Historical Probability of Major Drawdowns

Company	≥20%	≥30%	≥40%
TSLA	100.0%	80.0%	33.3%
NVDA	68.8%	50.0%	25.0%
AVGO	68.8%	25.0%	12.5%
AMZN	50.0%	18.8%	6.3%
TSM	50.0%	12.5%	6.3%
META	46.1%	30.8%	15.4%
AAPL	43.7%	25.0%	0.0%
GOOGL	43.7%	18.8%	6.3%
MSFT	25.0%	6.3%	0.0%

3.3 How to Interpret Historical Drawdown Recurrence

The Historical Drawdown Recurrence table measures how frequently a company experienced a given drawdown threshold **within a calendar year**.

The key point is that the analysis is based on **annual maximum drawdown**, not the number of independent corrections that occurred during the year.

For each complete calendar year, the stock receives one annual maximum drawdown observation.

For example, suppose a stock experiences:

- A 12% correction
- A recovery to a new high

- An 18% correction
- Another recovery
- A final 25% correction

The annual maximum drawdown is recorded as **25%**.

That year therefore qualifies as:

- a $\geq 10\%$ drawdown year;
- a $\geq 20\%$ drawdown year;

but not:

- a $\geq 30\%$ drawdown year.

Even though several separate corrections occurred, the year contributes only one annual maximum drawdown observation.

Recurrence Calculation

The recurrence interval is calculated as:

Observation Years $\div$ Number of Years Meeting the Drawdown Threshold

For example, NVIDIA has 16 complete calendar years in the main sample.

If NVIDIA experienced a drawdown of at least 30% in 8 of those 16 years:

$$16 \div 8 = 2.0 \text{ years}$$

The table therefore reports:

NVDA $\geq 30\%$ Drawdown: 2.0 years

The correct interpretation is:

Approximately one out of every two historical NVIDIA calendar years contained a drawdown of at least 30%.

It does not mean NVIDIA falls 30% exactly once every two years.

What Does “1.0 Year” Mean?

Tesla provides the clearest example.

The recurrence table shows:

Company	$\geq 20\%$ Drawdown Recurrence
TSLA	1.0 year
NVDA	1.45 years
AVGO	1.45 years
MSFT	4.0 years

Tesla has 15 complete calendar years in the main sample.

All 15 experienced at least a 20% annual maximum drawdown.

Therefore:

15 observation years $\div$ 15 qualifying years = 1.0 year

This means:

A 20% drawdown occurred in every complete Tesla calendar year included in the sample.

It does not mean:

- Tesla falls exactly 20% once every 12 months;
- Tesla experiences only one 20% correction each year;
- Tesla is guaranteed to fall 20% in the next calendar year.

It simply means that a 20% annual drawdown was historically universal within the observed Tesla sample.

NVIDIA and Broadcom

NVIDIA and Broadcom both show approximately:

$\geq 20\%$ Drawdown Recurrence = 1.45 years

Both have 16 complete observation years.

A recurrence interval of approximately 1.45 years corresponds to 11 qualifying years:

16 $\div$ 11 $\approx$ 1.45 years

Therefore, approximately 69% of their complete calendar years contained at least a 20% drawdown.

A 20% decline in NVIDIA or Broadcom is therefore relatively common.

This is particularly important for NVIDIA because its median annual maximum drawdown is already approximately 28.7%.

A 20% NVIDIA decline remains well within the stock's normal historical behavior.

Microsoft

Microsoft provides the opposite example.

Its $\geq 20\%$ recurrence interval is approximately four years.

Across 16 complete years, only four years experienced at least a 20% annual maximum drawdown.

Therefore:

$$16 \div 4 = 4 \text{ years}$$

Only approximately one quarter of Microsoft's complete calendar years experienced a 20% drawdown.

A 20% Microsoft decline is therefore substantially more unusual than a 20% decline in NVIDIA, Broadcom, or Tesla.

Using Recurrence as a Measure of Rarity

A useful interpretation framework is:

Recurrence Interval	Practical Interpretation
Around 1 year	Close to an annual norm
1–2 years	Frequent
2–3 years	Meaningful correction
3–6 years	Large drawdown
6–10 years	Rare drawdown
More than 10 years	Historical tail event
Not observed	Outside the observed annual sample

Examples:

Tesla $\geq 20\%$: 1.0 year

A 20% decline is close to normal historical behavior.

NVIDIA $\geq 30\%$: 2.0 years

A 30% drawdown is meaningful, but still relatively common.

Broadcom $\geq 40\%$: 8.0 years

A 40% drawdown is historically rare.

Microsoft $\geq 30\%$: 16 years

A 30% annual drawdown represents a historical tail event within the available sample.

This is why staged buying thresholds must differ materially by company.

3.4 How to Interpret Annual Historical Probability of Major Drawdowns

The Annual Historical Probability table presents the same underlying information from a different perspective.

Instead of asking:

How many years, on average, separate qualifying drawdown years?

it asks:

What percentage of complete calendar years experienced at least this level of drawdown?

The calculation is:

Annual Historical Probability = Qualifying Years $\div$ Complete Observation Years

For example, if 8 out of 16 years experienced at least a 30% drawdown:

8 $\div$ 16 = 50%

The annual historical probability is therefore 50%.

The corresponding recurrence interval is two years.

These are simply two ways of describing the same historical observation.

Tesla: Why $\geq 20\%$ Equals 100%

Tesla's $\geq 20\%$ annual historical probability is:

100%

This means every complete Tesla calendar year in the sample experienced an annual maximum drawdown of at least 20%.

It does not mean there is a mathematically certain 100% probability that Tesla must fall 20% next year.

It means only that:

15 out of 15 complete historical Tesla years crossed the 20% drawdown threshold.

Tesla remains extreme at deeper thresholds:

Threshold	Historical Annual Frequency
$\geq 20\%$	100.0%
$\geq 30\%$	80.0%
$\geq 40\%$	33.3%

The 30% figure is particularly important.

Approximately 80% of complete Tesla years experienced a drawdown of at least 30%.

Therefore, even a 30% Tesla decline has historically not been sufficient by itself to identify an unusually rare buying opportunity.

Only around the 40%–50% range does the historical frequency begin to decline materially.

NVIDIA: Large Drawdowns as Part of the Return Path

NVIDIA shows:

Threshold	Historical Annual Frequency
$\geq 20\%$	68.8%
$\geq 30\%$	50.0%
$\geq 40\%$	25.0%

Approximately two-thirds of complete NVIDIA years experienced at least a 20% drawdown.

Half experienced at least a 30% drawdown.

One quarter experienced at least a 40% drawdown.

This provides critical context for NVIDIA's extraordinary historical CAGR.

Its long-term compounding was not generated through a smooth return path.

Large corrections were a recurring feature of the investment experience.

The practical implication is:

- 20% is relatively normal;
- 30% begins to become more meaningful;
- 40% represents a much more important historical drawdown zone.

Broadcom: Frequent Corrections, Fewer Severe Extensions

Broadcom's $\geq 20\%$ frequency is identical to NVIDIA's:

68.8%

However, the deeper thresholds differ substantially.

Threshold	AVGO	NVDA
$\geq 20\%$	68.8%	68.8%
$\geq 30\%$	25.0%	50.0%
$\geq 40\%$	12.5%	25.0%

Broadcom frequently experienced 20% corrections, but those declines were substantially less likely than NVIDIA's to develop into 30%–40% drawdowns.

Historically, Broadcom therefore exhibited:

Frequent corrections, but fewer severe extensions of those corrections.

This helps explain why Broadcom generated an unusually strong combination of high CAGR and favorable drawdown characteristics.

For investment strategy purposes:

- A 20% Broadcom correction is common;
- A 30% correction is much more significant;
- A 40% drawdown is historically rare.

Microsoft: Lower Frequency Across Major Thresholds

Microsoft sits at the opposite end of the spectrum.

Threshold	Historical Annual Frequency
$\geq 20\%$	25.0%
$\geq 30\%$	6.3%
$\geq 40\%$	0.0% in the observed sample

Only one quarter of Microsoft's complete calendar years experienced at least a 20% drawdown.

A 30% annual drawdown occurred in only about 6% of the observed years.

No complete calendar year in the selected sample experienced a 40% annual maximum drawdown.

This does not mean a 40% Microsoft decline is impossible.

It means that such an event would lie outside the annual experience observed in the current dataset.

Therefore, Microsoft's 20%–30% drawdown zone carries substantially more statistical significance than the same percentage decline in NVIDIA or Tesla.

Apple and Alphabet: Moderate Drawdown Profiles

Apple:

Threshold	Historical Annual Frequency
$\geq 20\%$	43.7%
$\geq 30\%$	25.0%
$\geq 40\%$	0.0%

Alphabet:

Threshold	Historical Annual Frequency
$\geq 20\%$	43.7%
$\geq 30\%$	18.8%
$\geq 40\%$	6.3%

For both companies, 20% corrections are meaningful but not rare.

The more important threshold is around 30%.

A 30% decline occurs much less frequently and therefore provides a stronger historical signal for staged accumulation, assuming the business thesis remains intact.

Meta: Similar Normal Risk, Heavier Tail Risk

Meta's annual historical probabilities are:

Threshold	Historical Annual Frequency
$\geq 20\%$	46.1%
$\geq 30\%$	30.8%
$\geq 40\%$	15.4%

Its 20% frequency is relatively similar to Apple and Alphabet.

The difference becomes more visible at 30% and 40%.

Meta has historically experienced deeper annual drawdowns more frequently.

This supports its characterization as a company with:

Moderate normal-year risk but heavier tail risk.

For that reason, a 20%–30% Meta decline can support systematic accumulation, while a 40% or deeper drawdown should trigger increasingly detailed fundamental review.

Amazon and TSMC

Amazon:

- $\geq 20\%$: 50.0%

- $\geq 30\%$: 18.8%
- $\geq 40\%$: 6.3%

TSMC:

- $\geq 20\%$: 50.0%
- $\geq 30\%$: 12.5%
- $\geq 40\%$: 6.3%

For both stocks, a 20% drawdown is relatively common.

A 30% drawdown carries much more statistical significance.

TSMC is particularly interesting because the frequency falls from 50% at the 20% threshold to only 12.5% at the 30% threshold.

This suggests that many historical TSMC corrections stopped in the 20%–30% range rather than developing into much deeper declines.

This supports using approximately 30% as an important accumulation and review threshold, subject to semiconductor-cycle and geopolitical analysis.

Cross-Company Comparison

For a $\geq 20\%$ annual drawdown:

Company	Historical Frequency
TSLA	100.0%
NVDA	68.8%
AVGO	68.8%
AMZN	50.0%
TSM	50.0%
META	46.1%
AAPL	43.7%
GOOGL	43.7%
MSFT	25.0%

A 20% drawdown therefore represents:

- **Normal historical behavior for Tesla**

- **Frequent behavior for NVIDIA and Broadcom**
- **A meaningful but recurring correction for Amazon, TSMC, Meta, Apple, and Alphabet**
- **A relatively unusual event for Microsoft**

At the 30% threshold:

Company	Historical Frequency
TSLA	80.0%
NVDA	50.0%
META	30.8%
AAPL	25.0%
AVGO	25.0%
GOOGL	18.8%
AMZN	18.8%
TSM	12.5%
MSFT	6.3%

This is the central reason the investment framework uses different thresholds for different companies.

A 30% decline in Microsoft represents a historically unusual stress event.

A 30% decline in NVIDIA is a recurring feature of the stock's historical return path.

A 30% decline in Tesla is not particularly unusual by Tesla's own historical standards.

Important Statistical Limitation

These percentages are **historical sample frequencies**, not forward-looking probabilities.

For example:

TSLA $\geq 20\%$ = 100%

does not mean there is a guaranteed 100% probability that Tesla will experience another 20% drawdown next year.

It means:

Every complete Tesla calendar year included in the historical sample experienced at least one 20% annual drawdown.

Similarly:

MSFT $\geq 40\%$ = 0%

does not mean Microsoft cannot decline 40% in the future.

It means:

No complete calendar year in the selected historical sample produced a 40% annual maximum drawdown.

The sample contains only approximately 13–16 complete years depending on the company.

The results are therefore most useful for comparing the **relative drawdown behavior of the nine stocks**, rather than forecasting exact future probabilities.

4. Stock-by-Stock Strategy

4.1 Microsoft — Core DCA Compounder

Historical Profile

Metric	Historical Result
CAGR	21.0%
Positive Years	81.3%
Median Annual Max Drawdown	-16.0%
Annualized Volatility	23.1%
Downside Volatility	16.7%
Typical Worst 1M	-10.6%
Typical Worst 3M	-11.4%
Median Recovery	80.5 days
$\geq 20\%$ Drawdown	~4 years
$\geq 30\%$ Drawdown	~16 years
$\geq 40\%$ Drawdown	Not observed

Historical Interpretation

Microsoft has historically offered the strongest combination of business stability and price stability among the companies in this study.

Its typical annual maximum drawdown is only around 16%, and a 20% decline has been relatively uncommon.

Waiting indefinitely for a 30% Microsoft drawdown is therefore not an efficient core strategy.

The better approach is to remain continuously invested and accelerate purchases when the stock enters unusually deep drawdowns.

Accumulation Framework

0% to -10%

Normal volatility.

Continue regular DCA.

Around -15%

Still broadly consistent with a normal year.

A modest increase is reasonable if valuation has improved, but this is not a rare opportunity.

Around -20%

First major accumulation zone.

Historically, this occurred approximately once every four years.

A reasonable response is to increase toward approximately 2× the normal contribution.

Around -25%

A deeper-than-normal correction.

If the decline is primarily macro-driven rather than company-specific, expected long-term returns may have improved materially.

Around -30%

Historically unusual.

If Azure, Microsoft 365, Security, AI monetization, margins, and free cash flow remain intact, this can justify approximately 3× normal staged deployment.

Thesis Review

Monitor:

- Azure growth
- Azure market position
- Microsoft 365 retention
- Microsoft 365 pricing power
- Copilot monetization
- Security growth
- Operating margin
- Free cash flow
- AI infrastructure capital intensity
- Return on AI CapEx

Recommended Strategy

Continuous DCA; accelerate near -20%; become aggressive near -30% if the business thesis remains intact.

4.2 Alphabet — Core DCA with Drawdown Acceleration

Historical Profile

Metric	Historical Result
CAGR	20.7%
Positive Years	75.0%
Median Annual Max Drawdown	-19.1%
Annualized Volatility	28.0%
Downside Volatility	18.4%
Median Recovery	121.5 days
≥20% Drawdown	~2.29 years
≥30% Drawdown	~5.33 years
≥40% Drawdown	Rare

Accumulation Framework

0% to -10%

Normal volatility.

Around -20%

Meaningful but not rare.

Increase toward approximately 1.5× base.

Around -30%

Major accumulation zone.

Increase toward approximately 2.5× base if Search, Cloud, YouTube, and AI economics remain intact.

Around -40%

Tail event.

Require a deeper review of whether AI is structurally altering Search economics before aggressive deployment.

Thesis Review

Monitor:

- Search query trends
- Search revenue growth
- Search advertising pricing
- AI Search monetization
- Gemini adoption
- Google Cloud growth
- Cloud operating margin
- YouTube advertising
- YouTube subscriptions
- CapEx
- Free cash flow

Recommended Strategy

Maintain continuous DCA; increase around -20%; become substantially more aggressive around -30%; require deeper thesis validation near -40%.

4.3 Apple — Core Quality Compounder

Historical Profile

Metric	Historical Result
CAGR	26.5%
Positive Years	81.3%
Median Annual Max Drawdown	-18.8%
Annualized Volatility	26.5%
Downside Volatility	19.0%
Median Recovery	100.5 days
≥20% Drawdown	~2.29 years
≥30% Drawdown	~4 years
≥40% Drawdown	Not observed

Accumulation Framework

0% to -10%

Normal.

Around -20%

Begin increasing toward approximately 1.5× base.

Around -30%

Major accumulation zone.

Approximately 2.5× base may be appropriate if ecosystem retention, Services growth, and cash generation remain strong.

Approaching -40%

Outside the observed annual sample.

Require a full review before further increasing exposure.

Thesis Review

Monitor:

- Active installed base
- iPhone replacement cycle
- Services growth
- Services gross margin
- Ecosystem retention
- China exposure
- AI competitiveness
- Free cash flow
- Valuation

Recommended Strategy

Continuous DCA; increase at -20%; allocate more aggressively near -30%; review the thesis if the decline becomes historically unprecedented.

4.4 Broadcom — High-Return Compounder with Dry Powder

Historical Profile

Metric	Historical Result
CAGR	41.9%
Positive Years	93.8%
Median Annual Max Drawdown	-22.9%
Annualized Volatility	34.7%
Downside Volatility	24.6%
Median Recovery	89.5 days
≥20% Drawdown	~1.45 years
≥30% Drawdown	~4 years
≥40% Drawdown	~8 years

Accumulation Framework

Around -20%

Relatively common.

Only a modest increase, perhaps around 1.25× base.

Around -30%

Much more meaningful.

Approximately 2× base becomes more attractive.

Around -40%

Historically rare.

If AI networking, custom silicon, VMware economics, and free cash flow remain healthy, this can justify approximately 3× staged deployment.

Thesis Review

Monitor:

- AI semiconductor revenue
- AI networking
- Custom accelerators
- Hyperscaler concentration
- VMware
- Debt
- Free cash flow
- AI CapEx cycle

Capital Allocation Approach

A practical structure could allocate:

- 70%–80% of planned Broadcom investment through normal DCA
- 20%–30% as dry powder for 30%–40% corrections

Recommended Strategy

Stay invested, treat -20% as relatively normal, increase materially near -30%, and use -40% as a major opportunity only if the cycle remains intact.

4.5 Meta — Platform Compounder with Tail-Risk Discipline

Historical Profile

Metric	Historical Result
CAGR	28.1%
Positive Years	84.6%
Median Annual Max Drawdown	-19.7%
Annualized Volatility	36.5%
Downside Volatility	24.3%
Median Recovery	78 days
≥20% Drawdown	~2.17 years
≥30% Drawdown	~3.25 years
≥40% Drawdown	~6.5 years
≥50% Drawdown	~13 years

Accumulation Framework

Around -20%

Meaningful but not rare.

Increase toward approximately 1.5× base.

Around -30%

Substantial correction.

Approximately 2× base can be considered if engagement and advertising economics remain healthy.

Around -40%

Statistically attractive but fundamentally more dangerous.

A 2.5×–3× allocation should only follow a full Thesis Review.

Around -50% or deeper

Suspend mechanical averaging down and rebuild the investment thesis.

Thesis Review

Monitor:

- Instagram engagement
- Reels engagement
- Facebook usage
- WhatsApp monetization
- Advertising impressions
- Advertising pricing
- Advertiser ROI
- AI recommendation effectiveness
- Reality Labs losses
- CapEx
- Operating margin
- Free cash flow

Recommended Strategy

Use systematic buying at -20% and -30%, but require mandatory thesis validation at -40% and deeper.

4.6 Amazon — Growth Platform with Large-Correction Upsizing

Historical Profile

Metric	Historical Result
CAGR	24.7%
Positive Years	81.3%
Median Annual Max Drawdown	-21.2%
Annualized Volatility	32.4%
Downside Volatility	21.2%
Median Recovery	107 days
≥20% Drawdown	~2 years
≥30% Drawdown	~5.33 years
≥40% Drawdown	Rare

Accumulation Framework

Around -20%

Meaningful but relatively common.

Approximately 1.5× base.

Around -30%

Major opportunity zone.

Approximately 2.5× base if AWS, advertising, margins, and FCF remain intact.

Around -40%

Rare historical event.

Potentially approximately 3× base, but only after confirming AWS economics and capital efficiency.

Thesis Review

Monitor:

- AWS growth
- AWS operating margin
- Generative AI monetization
- Advertising
- Retail margin
- International profitability
- CapEx
- Operating cash flow
- Free cash flow

Recommended Strategy

Maintain long-term exposure, add moderately around -20%, and reserve major incremental capital for -30% and deeper corrections.

4.7 TSMC — Cyclical Compounder

Historical Profile

Metric	Historical Result
CAGR	26.4%
Positive Years	87.5%
Median Annual Max Drawdown	-20.3%

Metric	Historical Result
Annualized Volatility	26.8%
Downside Volatility	19.0%
Median Recovery	130.5 days
≥20% Drawdown	~2 years
≥30% Drawdown	~8 years
≥40% Drawdown	Rare

Accumulation Framework

Around -20%

Normal semiconductor correction.

Approximately 1.5× base can be considered.

Around -30%

Historically much more unusual.

Approximately 2.5×–3× can be considered after reviewing the semiconductor cycle.

Around -40% or deeper

Full fundamental and geopolitical review.

Thesis Review

Monitor:

- Leading-edge utilization
- Advanced-node demand
- N2 execution
- AI accelerator demand
- Smartphone and PC cycle
- Customer inventory
- Wafer pricing
- Gross margin
- CapEx
- Customer concentration
- Taiwan geopolitical risk

Recommended Strategy

Maintain a smaller base DCA; use -20% as an initial add zone; treat -30% as a major opportunity only after cycle analysis.

4.8 NVIDIA — High-Beta AI Compounder

Historical Profile

Metric	Historical Result
CAGR	46.2%
Positive Years	68.8%
Median Annual Max Drawdown	-28.7%
Annualized Volatility	46.0%
Downside Volatility	27.9%
Median Recovery	89.5 days
≥20% Drawdown	~1.45 years
≥30% Drawdown	~2 years
≥40% Drawdown	~4 years
≥50% Drawdown	~4 years

Accumulation Framework

0% to -10%

Noise.

Around -20%

Normal high-beta correction.

Do not deploy major dry powder solely because of the price decline.

Around -30%

First genuinely meaningful accumulation zone.

Approximately 1.5×–2× base.

Around -40%

Major accumulation zone.

Approximately 2.5×–3× base if the AI investment thesis remains intact.

Around -50%

Severe stress.

Requires a full cycle and competitive review.

Thesis Review

Monitor:

- Hyperscaler AI CapEx
- GPU demand
- GPU utilization
- CUDA ecosystem
- Custom ASIC competition
- AMD
- HBM supply
- Networking
- Data Center growth
- Gross margin
- Customer concentration
- AI infrastructure ROI

Recommended Strategy

Maintain a smaller base allocation, avoid overreacting to 20% corrections, increase around -30%, and reserve the most aggressive capital deployment for -40% or deeper drawdowns.

4.9 Tesla — High-Beta Optionality Asset

Historical Profile

Metric	Historical Result
CAGR	44.6%
Positive Years	86.7%
Median Annual Max Drawdown	-35.7%

Metric	Historical Result
Annualized Volatility	54.1%
Downside Volatility	36.5%
Typical Worst 1M	-27.0%
Typical Worst 3M	-24.4%
Median Recovery	204 days
≥20% Drawdown	~1 year
≥30% Drawdown	~1.25 years
≥40% Drawdown	~3 years
≥50% Drawdown	~7.5 years

Accumulation Framework

Around -20%

Normal Tesla volatility.

Do not increase materially based on drawdown alone.

Around -30%

Still relatively common.

Maintain only normal or modest deployment.

Around -40%

Begins to become meaningfully unusual.

Approximately 1.5×–2× base can be considered if the long-term thesis remains intact.

Around -50%

Materially rarer.

Approximately 2.5× base may be justified only after detailed thesis review.

Around -60% or deeper

Suspend the mechanical drawdown framework.

Rebuild the valuation and investment thesis from the ground up.

Thesis Review

Monitor:

- Vehicle deliveries
- Automotive revenue
- Automotive gross margin
- Pricing
- FSD adoption
- FSD economics
- Robotaxi
- Regulatory progress
- Optimus
- Energy storage
- Free cash flow
- CapEx
- Balance sheet
- Valuation of future businesses

Recommended Strategy

Maintain only a small base allocation, avoid aggressive buying after ordinary 20%–30% declines, and reserve major incremental capital for 40%–50% drawdowns after fundamental validation.

4.10 Stock-by-Stock Strategy Summary

Company	Normal Approach	First Meaningful Add	Major Add	Deep Drawdown Treatment
MSFT	Continuous DCA	-20%	-30%	Thesis review beyond historical extremes
GOOGL	Continuous DCA	-20%	-30%	Strong accumulation near -40% only if Search thesis remains intact
AAPL	Continuous DCA	-20%	-30%	Full review near historically unprecedented levels
AVGO	DCA + Dry Powder	Small add near -20%	-30%	Major add near -40% if AI/cycle thesis remains intact
META	Moderate DCA	-20%	-30%	Mandatory thesis review near -40%
AMZN	Continuous DCA	-20%	-30%	Large add near -40% if AWS and FCF remain healthy

Company	Normal Approach	First Meaningful Add	Major Add	Deep Drawdown Treatment
TSM	Light DCA	-20%	-30%	Semiconductor-cycle and geopolitical review
NVDA	Light DCA	-30%	-40%	Full cycle review near -50%
TSLA	Minimal Base	-40%	-50%	Full valuation and thesis reassessment near -60%

5. Portfolio Implementation

5.1 Core Compounders

The first portfolio bucket consists of:

- Microsoft
- Alphabet
- Apple

These companies are best treated as long-duration compounding assets.

The primary objective is to avoid excessive market timing.

Investors should:

- Maintain ongoing exposure
- Use regular DCA
- Avoid holding excessive cash while waiting for extremely rare corrections
- Increase capital deployment when 20%–30% drawdowns materially improve expected returns

These companies should generally receive the highest baseline DCA intensity.

5.2 Opportunistic Compounders

The second bucket consists of:

- Broadcom
- Meta
- Amazon

These companies combine strong long-term economics with greater drawdown variability.

The optimal approach is:

- Maintain meaningful baseline positions
- Continue regular investment
- Reserve part of the annual allocation as dry powder
- Increase purchases more aggressively when the stock enters statistically unusual drawdown ranges

Broadcom is the strongest example of a high-return compounder.

Meta requires more tail-risk discipline.

Amazon becomes increasingly attractive when corrections move beyond its normal 20% range.

5.3 Cyclical Compounder

TSMC occupies its own category.

Its long-term return and normal volatility are attractive enough to justify ongoing ownership.

However, staged buying should incorporate the semiconductor cycle.

Investors should distinguish between:

Temporary cyclical weakness

and

Structural impairment or geopolitical repricing

A 30% TSMC decline may be historically attractive, but the probability of value creation depends heavily on why the decline occurred.

5.4 High-Beta and Optionality Bucket

NVIDIA and Tesla should receive lower baseline DCA intensity.

The reason is not that they are lower-quality businesses.

The reason is that their normal volatility is much higher.

If an investor uses the same 20% dip-buying rule for Microsoft and NVIDIA, the NVIDIA allocation may grow far too quickly during ordinary volatility.

For NVIDIA:

- 20% is largely normal
- 30% begins to matter
- 40% is a major accumulation zone
- 50% requires a full cycle review

For Tesla:

- 20% is normal
- 30% remains common
- 40% begins to matter
- 50% is materially more interesting
- 60% requires rebuilding the thesis

5.5 Suggested Drawdown Allocation Matrix

Assume a normal investment contribution is defined as **1× Base DCA**.

Company	Base DCA	-20%	-30%	-40%	-50%	Strategy Type
MSFT	1×	2×	3×	Thesis Review	—	Core DCA
GOOGL	1×	1.5×	2.5×	3× *	—	Core DCA
AAPL	1×	1.5×	2.5×	Thesis Review	—	Core DCA
AVGO	0.75–1×	1.25×	2×	3× *	—	Opportunistic
META	0.75–1×	1.5×	2×	2.5–3× *	Thesis Review	Opportunistic
AMZN	0.75–1×	1.5×	2.5×	3× *	Thesis Review	Opportunistic
TSM	0.5–0.75×	1.5×	2.5–3× *	Thesis Review	Thesis Review	Cyclical
NVDA	0.5–0.75×	1×	1.5–2×	2.5–3× *	3× *	High Beta
TSLA	0–0.5×	No incremental add	1×	1.5–2× *	2.5× *	Optionality

* Requires confirmation that the long-term investment thesis remains intact.

The multiplier represents the pace of incremental investment rather than a target portfolio weight.

If the standard monthly contribution to a stock is \$1,000:

- $1\times = \$1,000$
- $1.5\times = \$1,500$
- $2\times = \$2,000$
- $3\times = \$3,000$

The goal is not to identify the exact bottom.

Instead, investment intensity rises as the drawdown becomes increasingly unusual relative to the company's own historical distribution.

5.6 Thesis and Valuation Overlay

Drawdown magnitude should never be used alone.

A more complete investment framework evaluates four conditions.

Business Quality

Does the company still possess durable competitive advantages?

Thesis Intact

Has the original reason for owning the company remained valid?

Drawdown Rarity

Is the current decline unusually deep relative to the company's historical distribution?

Valuation

Has the decline materially improved expected long-term returns?

The strongest buying opportunities occur when all four conditions align.

A deep drawdown caused by market fear is fundamentally different from a deep drawdown caused by permanent impairment.

5.7 Portfolio-Level Dry Powder

Dry powder should not be distributed equally across all stocks.

Lower Dry-Powder Requirement

- MSFT
- GOOGL
- AAPL

Because deep corrections are less frequent, holding too much cash may reduce long-term compounding.

Moderate Dry-Powder Requirement

- AVGO
- META
- AMZN
- TSM

These companies provide more frequent opportunities to increase exposure during larger corrections.

Higher Dry-Powder Requirement

- NVDA
- TSLA

Because 20%–30% declines occur relatively frequently, investors should avoid deploying all available capital too early.

More capital should be reserved for deeper corrections.

5.8 Statistical Limitations

This framework should not be interpreted as a deterministic trading model.

The primary sample covers approximately 2010–2025, which remains a relatively short period for estimating true tail probabilities.

If a 40% drawdown occurred once in 16 years, it does not mean future 40% drawdowns will occur exactly once every 16 years.

It simply means that the event was rare within the observed sample.

Similarly, “not observed” does not mean impossible.

For practical purposes:

- One to three years indicates relatively frequent behavior
- Three to six years indicates a meaningful large drawdown
- Six to ten years indicates a rare event

- More than ten years indicates a tail event
-

5.9 Annual Drawdown vs Independent Drawdown Events

The current analysis uses annual maximum drawdown.

This is useful because it answers:

How likely is a stock to experience a given level of drawdown within a calendar year?

However, each calendar year can contribute only one annual maximum drawdown observation.

A future extension should identify independent drawdown events using daily price data.

That would allow the analysis to answer:

- How many independent 10% corrections occur per year?
- How frequently does a stock experience a 20% correction?
- What is the median time between independent drawdowns?
- How long does a typical correction last?
- How quickly does the stock recover?
- Do drawdowns cluster during specific market regimes?

This would create a more complete event-based drawdown model.

5.10 Final Portfolio Takeaways

The nine companies should not be managed using the same investment rule.

Microsoft, Alphabet, and Apple

These are the strongest candidates for continuous long-term DCA.

Their normal drawdowns are moderate and deep corrections are infrequent enough that waiting for a crash can create significant opportunity cost.

Preferred strategy:

Stay invested and accelerate purchases during 20%–30% drawdowns.

Broadcom, Meta, and Amazon

These are better suited to a hybrid approach.

Preferred strategy:

Maintain baseline exposure while reserving capital for 30%–40% corrections.

TSMC

TSMC should be treated as a cyclical compounder.

Preferred strategy:

Maintain long-term exposure while combining price-based signals with semiconductor-cycle analysis.

NVIDIA

NVIDIA's historical compounding has been exceptional, but its high-beta characteristics remain important.

Preferred strategy:

Maintain a smaller base allocation, avoid overreacting to 20% corrections, and become progressively more aggressive in the 30%–40% range.

Tesla

Tesla requires the most conservative use of a mechanical dip-buying framework.

Preferred strategy:

Maintain limited base exposure, treat 20%–30% declines as ordinary volatility, and reserve major incremental capital for 40%–50% drawdowns after validating the fundamental thesis.

Conclusion

The purpose of this framework is not to forecast market bottoms.

It is to improve capital-allocation discipline.

The most common mistake in long-term growth investing is to treat all declines as equivalent.

They are not.

A 20% decline in Microsoft carries different information from a 20% decline in NVIDIA.

A 30% decline in Alphabet carries different information from a 30% decline in Tesla.

The historical distribution of each stock determines how unusual a given decline actually is.

The most robust long-term strategy is therefore not:

Buy more simply because the stock has fallen.

It is:

Maintain exposure to high-quality compounders, understand the normal drawdown range of each company, increase capital deployment when a drawdown becomes unusually deep relative to its own history, and verify that the fundamental thesis and valuation remain attractive before committing major incremental capital.

This approach converts market volatility from an emotional event into a structured capital-allocation decision.